

# Clear figures, stronger stories

Scientific workflows: Tools and Tips 

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2025-04-17

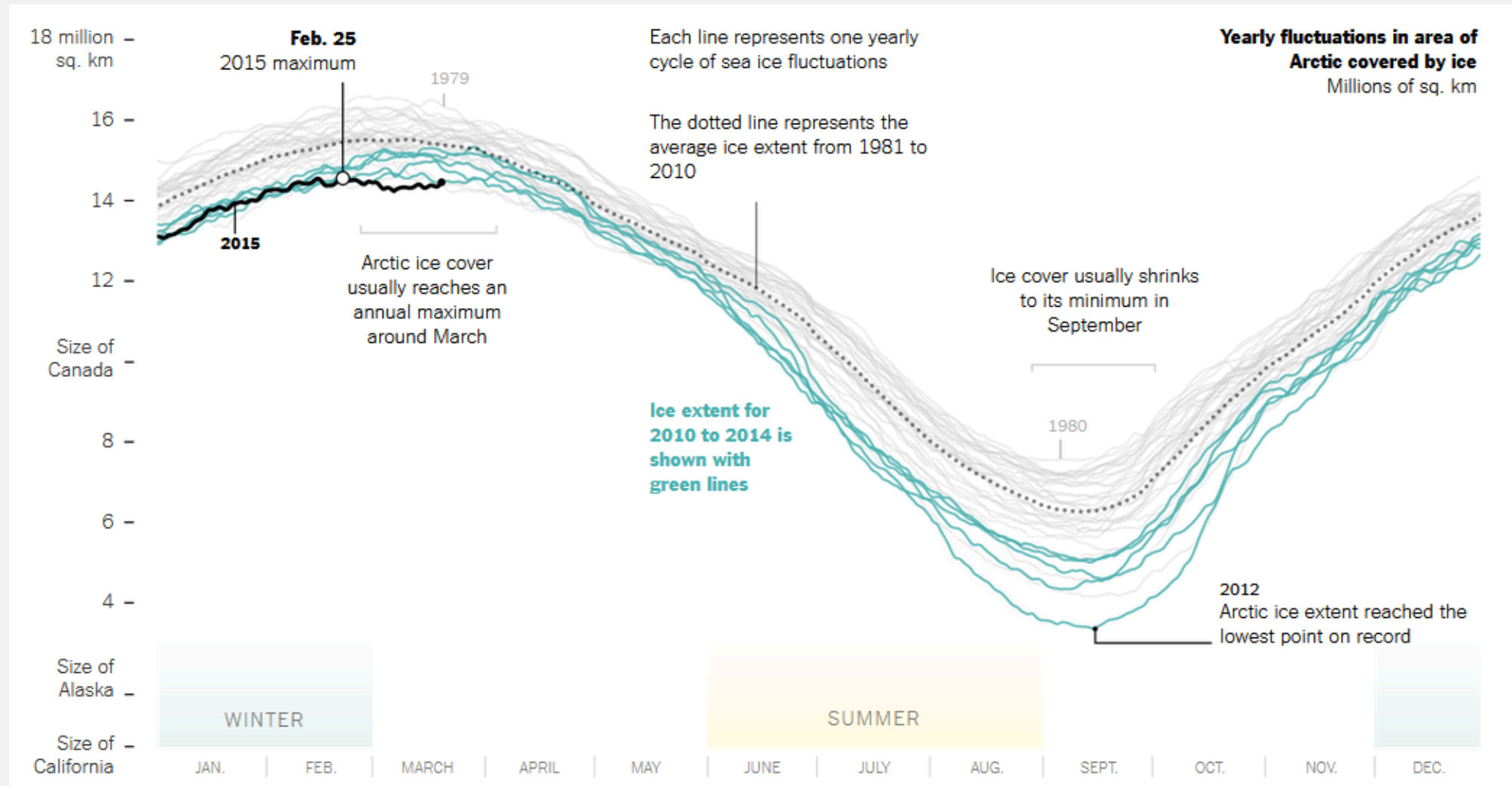
# What is this lecture series?

## Scientific workflows: Tools and Tips

 Every 3rd Thursday  4-5 p.m.  Webex

- One topic from the world of scientific workflows
- Material provided [online](#)
- If you don't want to miss a lecture
  - [Subscribe to the mailing list](#)

# Motivation



Annual changes in Arctic sea ice cover by [Derek Watkins \(New York Times\)](#)

# What makes a good figure?

- **Correct and transparent**  
Truthful representation of the data, data integrity
- **Useful**  
Supports the main point you want to make
- **Easy to read and understand**  
Accessible for everyone
- **Beautiful**  
Visually interesting and pleasing
- **Appropriate**  
Different outlets have different requirements/freedoms

# 7 steps for better figures



# 1: Consider the context

# Consider the context

- **Who** is your **audience**?  
Familiarity with the topic
- **What** are **common practices** in your field?  
Established plot types, colors, ...
- **Where** do you present your figure?  
Different contexts require different designs

# Contexts in science

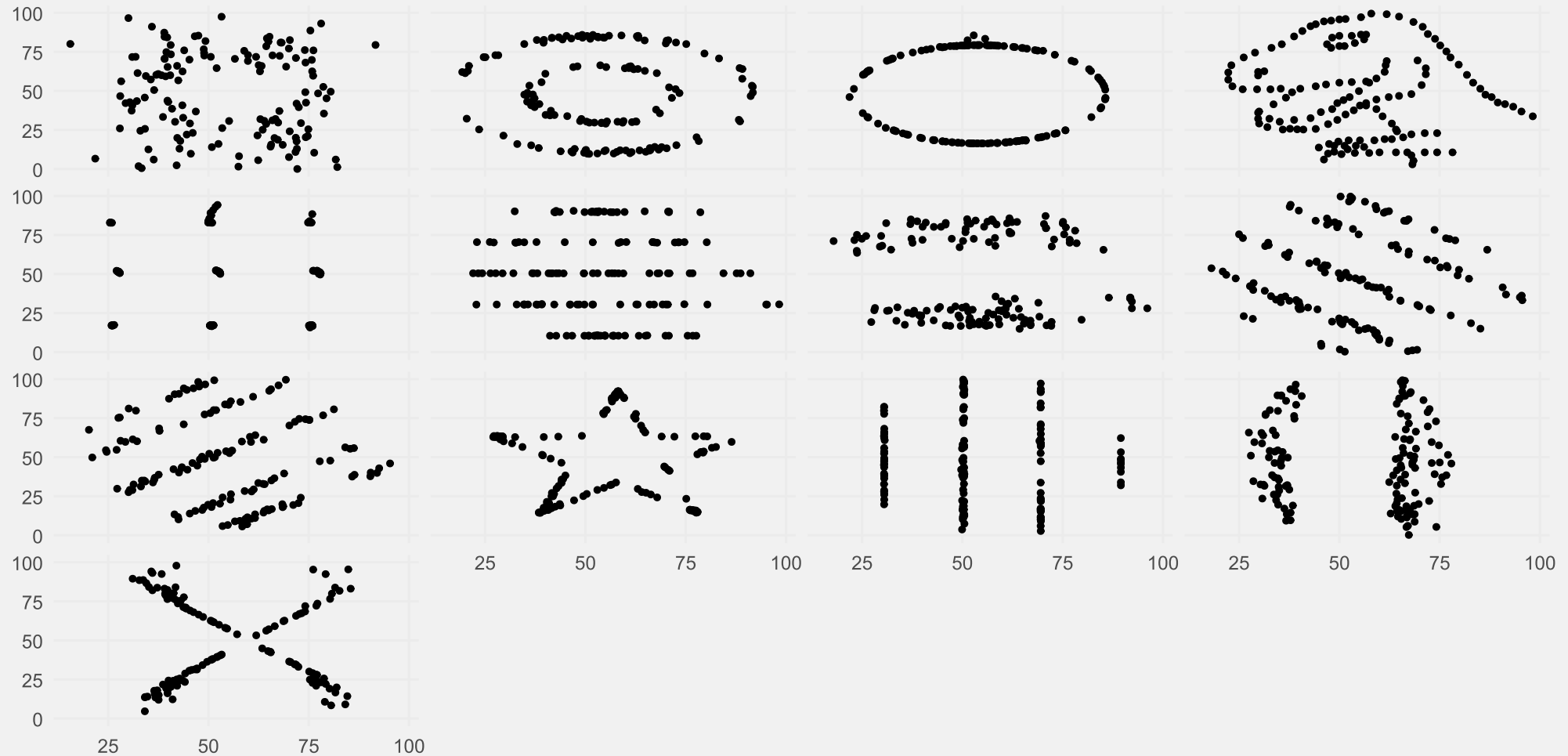
| Context | Things to consider                                                                                                                                                                                                   |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Paper   | <ul style="list-style-type: none"><li>- Journal requirements</li><li>- Usually read on PC but also print (B/W)</li><li>- More time → Higher complexity</li></ul>                                                     |
| Poster  | <ul style="list-style-type: none"><li>- More open design choice</li><li>- Attract people from far</li><li>- You quickly loose people to other posters</li><li>- Medium complexity (depending on the event)</li></ul> |
| Talk    | <ul style="list-style-type: none"><li>- You can use animations to guide through</li><li>- Little time → less complex</li></ul>                                                                                       |



# 2: Make your data transparent

# Don't hide data behind summaries

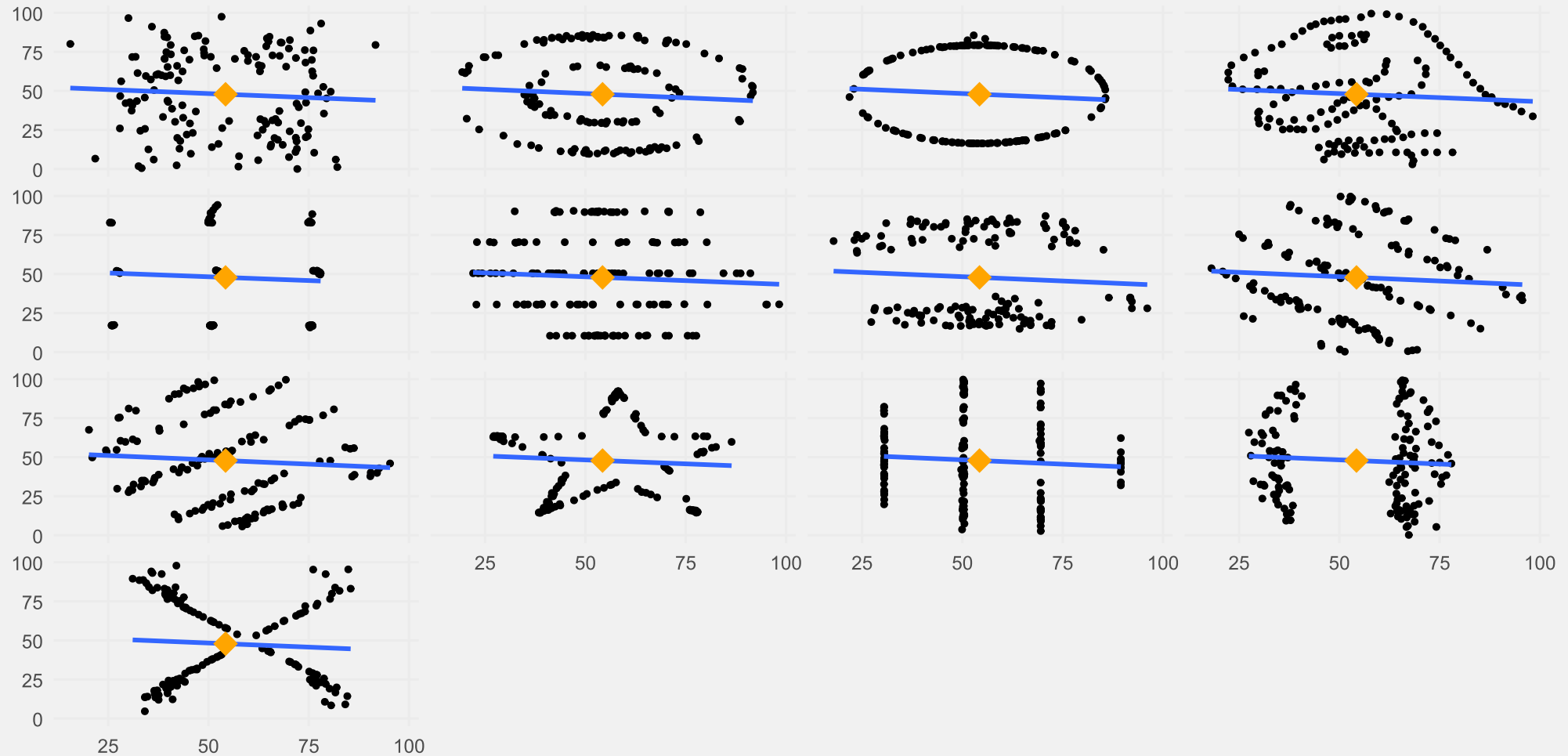
Datasaurus Dozen: Different distributions, same statistics



Data from the `datasauRus` package based on Matejka, J., & Fitzmaurice, G. (2017). Same Stats, Different Graphs.

# Don't hide data behind summaries

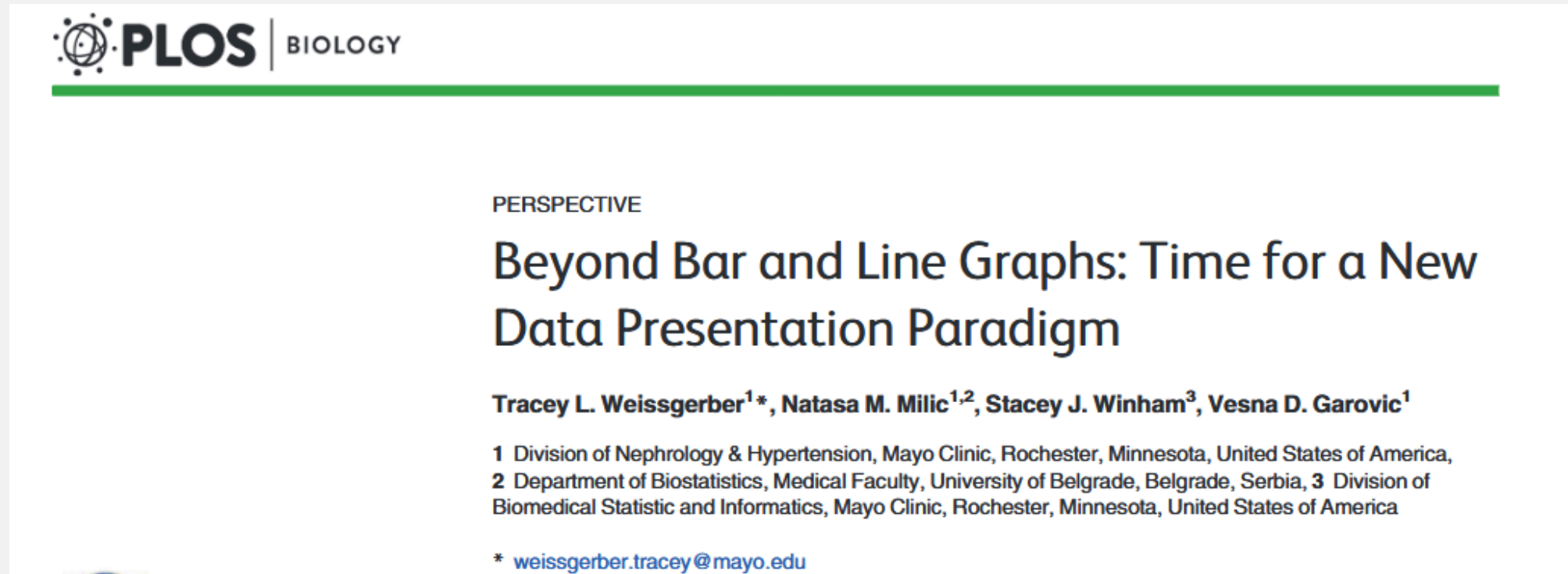
Datasaurus Dozen: Different distributions, same statistics



Data from the `datasauRus` package based on Matejka, J., & Fitzmaurice, G. (2017). Same Stats, Different Graphs.

# Don't hide data behind summaries

Bar graphs hide a lot of information about the data.



<https://doi.org/10.1371/journal.pbio.1002128>

# Don't hide data behind summaries

Same bar plot - different data & statistical test results

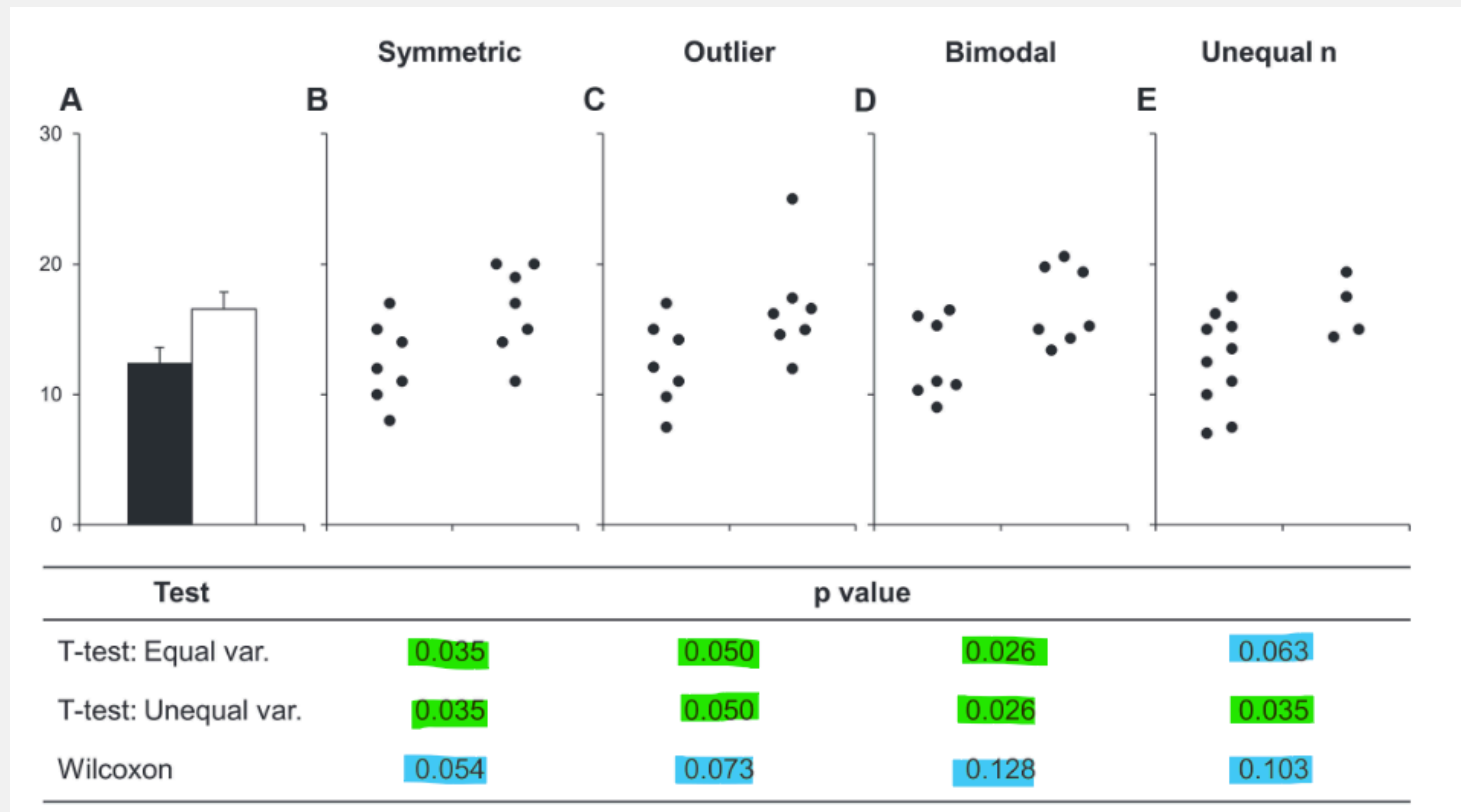
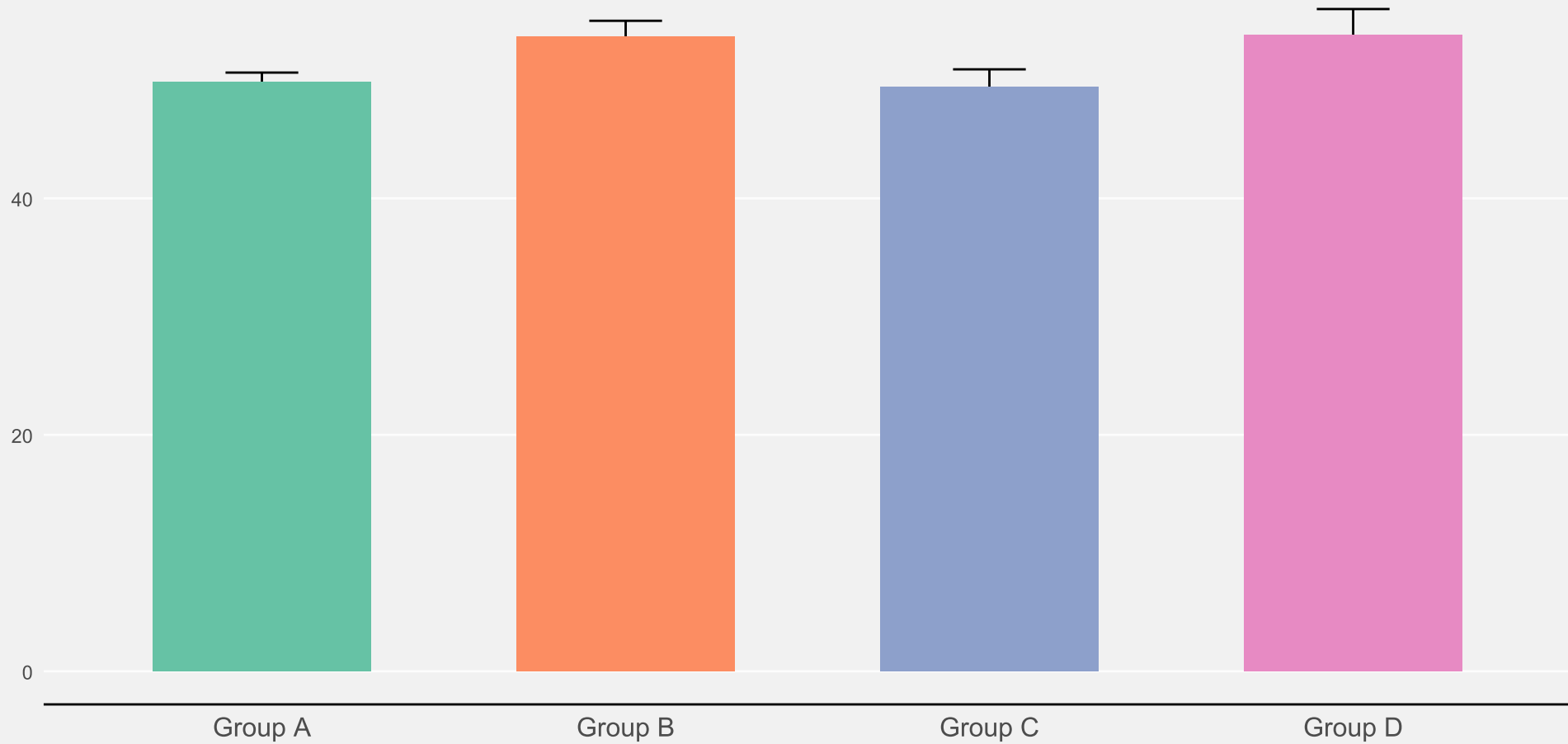


Figure 1 from Weissgerber et al. 2015

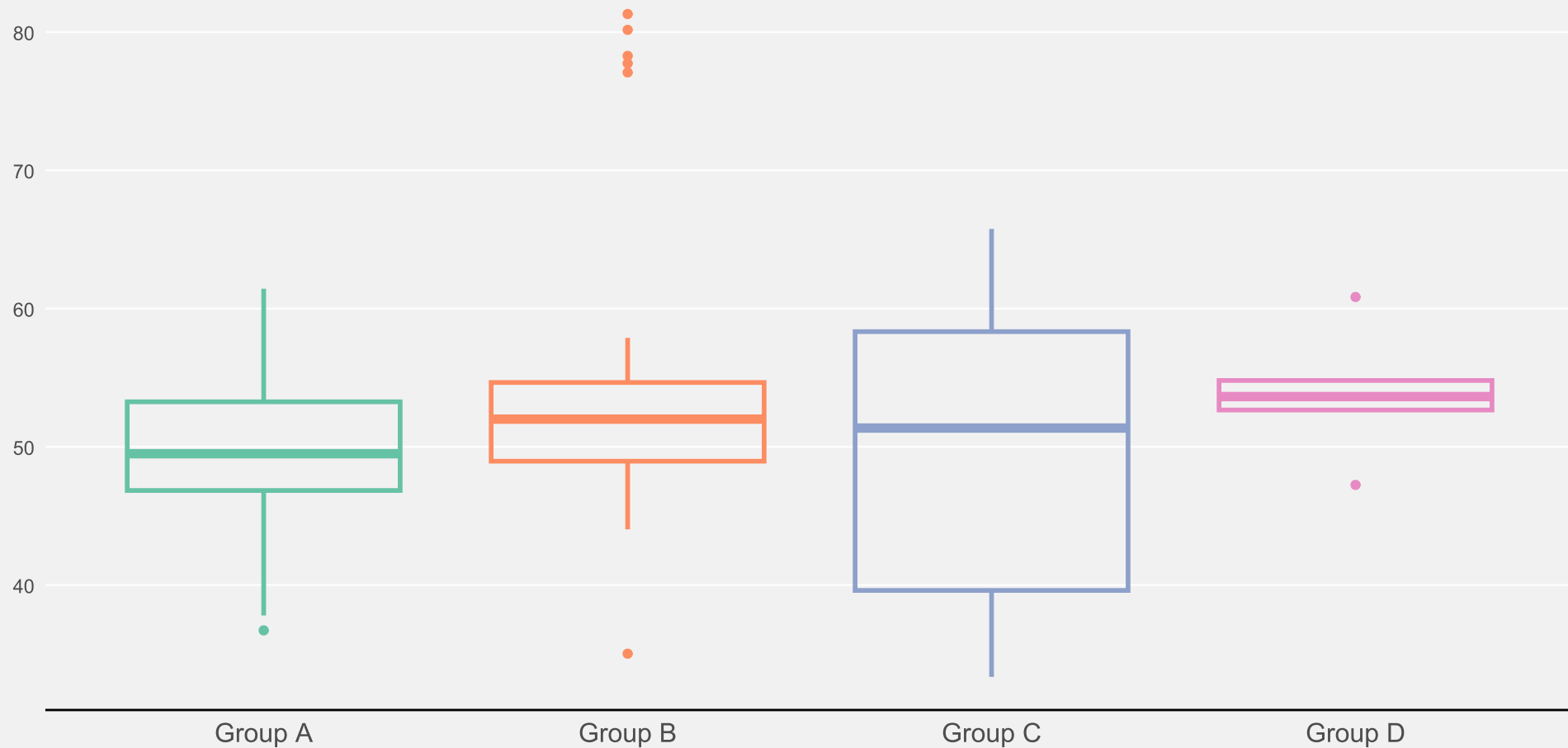
# Alternatives to bar plots

Bar plots only show mean  $\pm$  SE/SD.



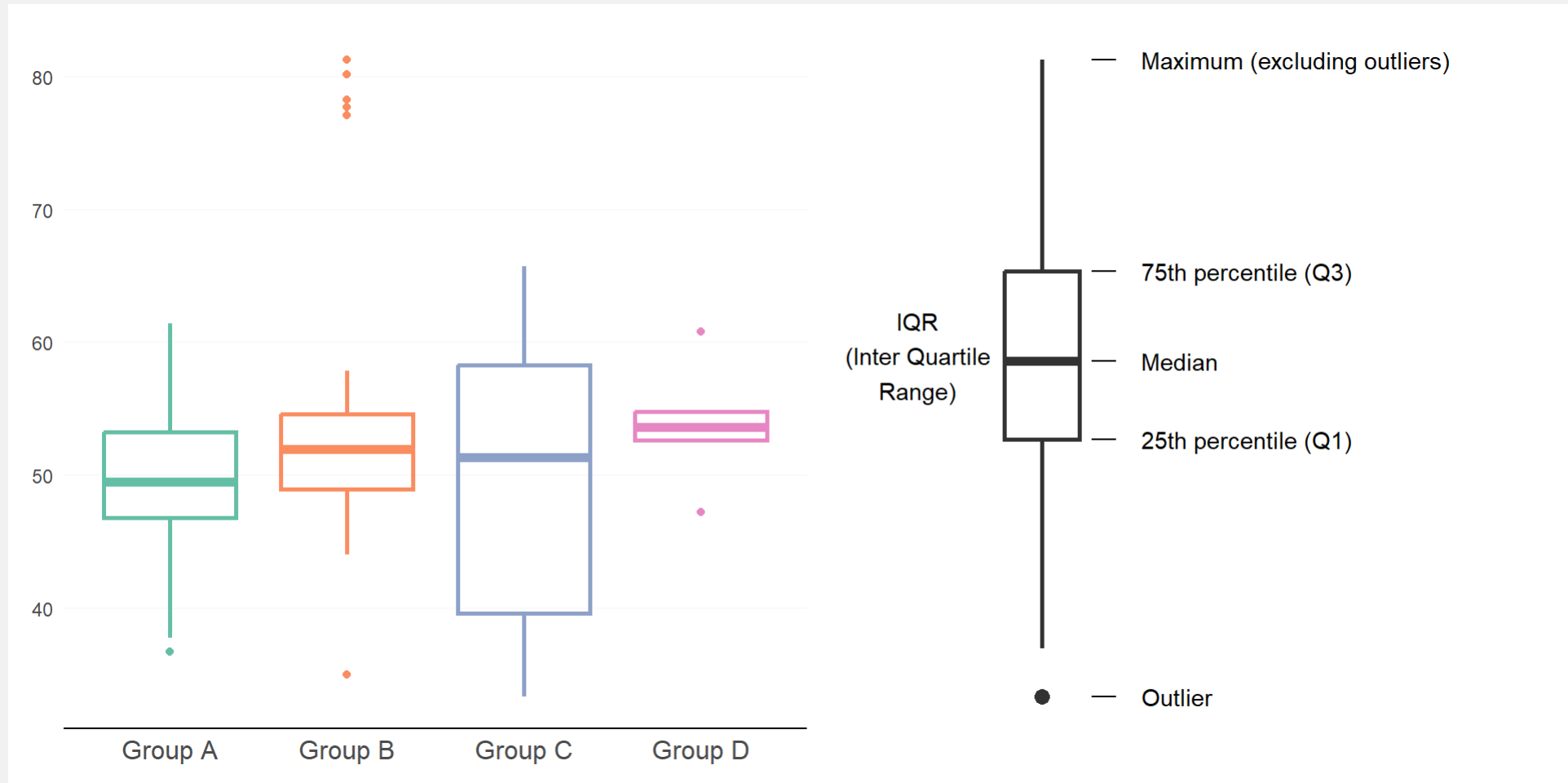
# Alternatives to bar plots

A box plot is already better (shows more of the distribution)



# Alternatives to bar plots

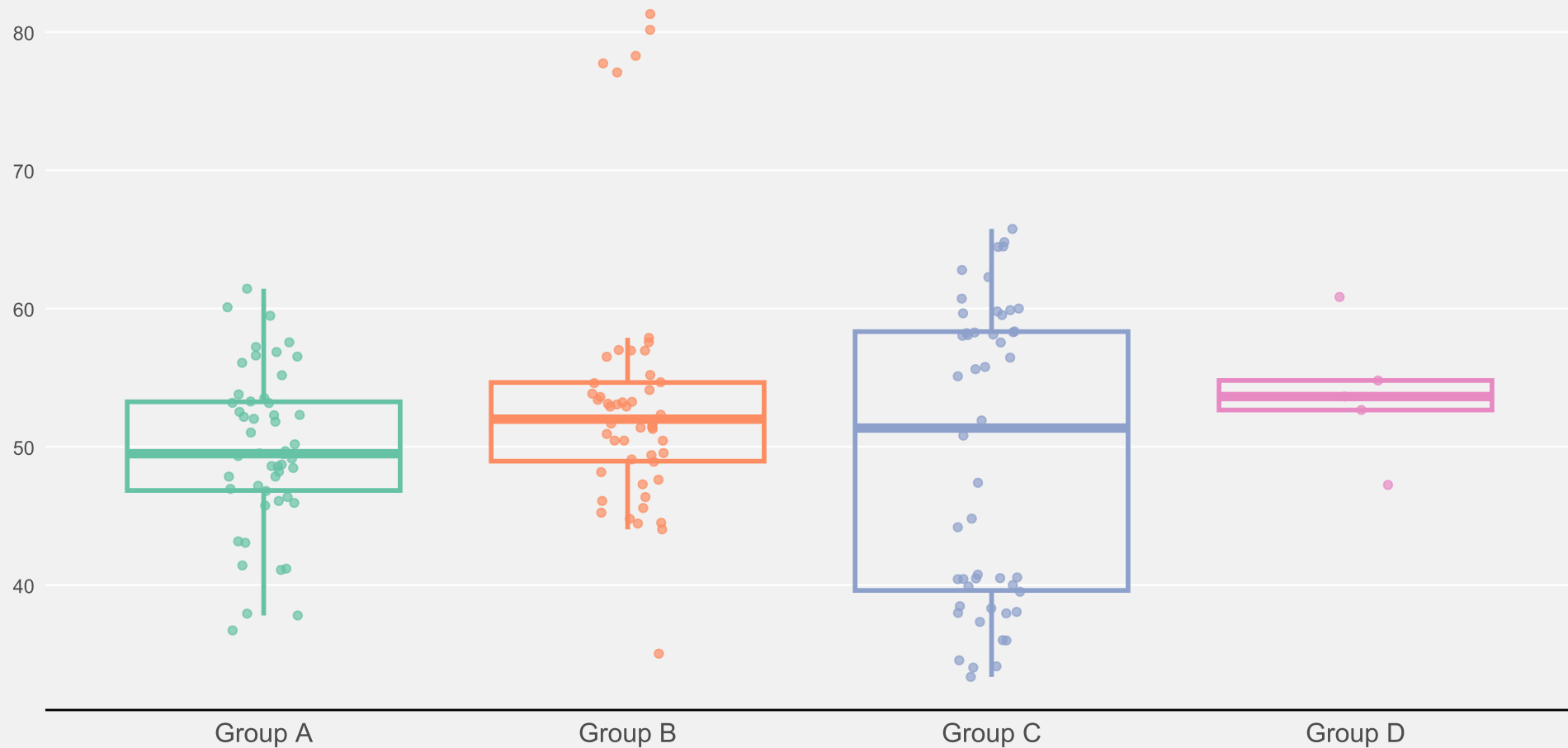
A box plot is already better (shows more of the distribution)





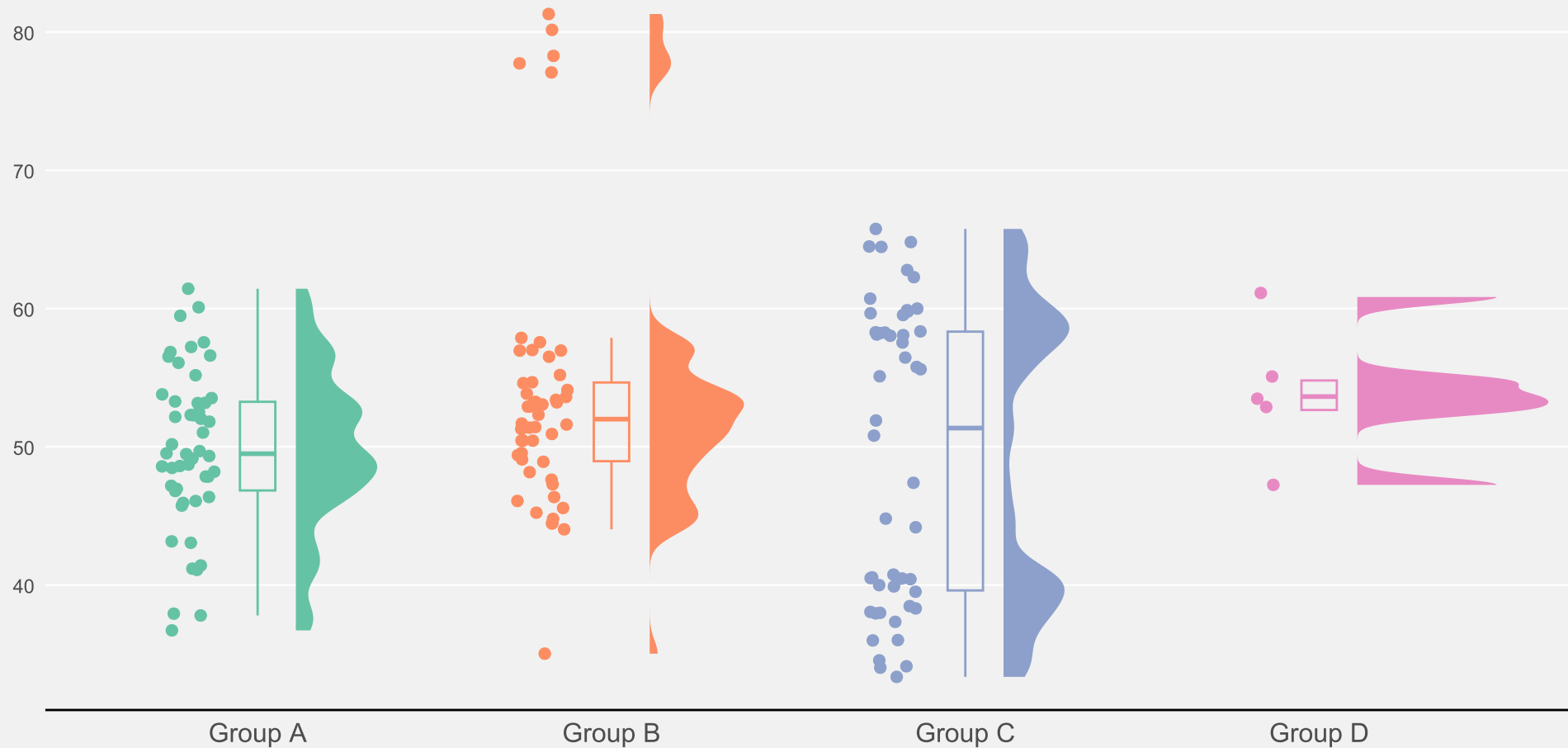
# Alternatives to bar plots

Add raw data points to increase the information content of the plot



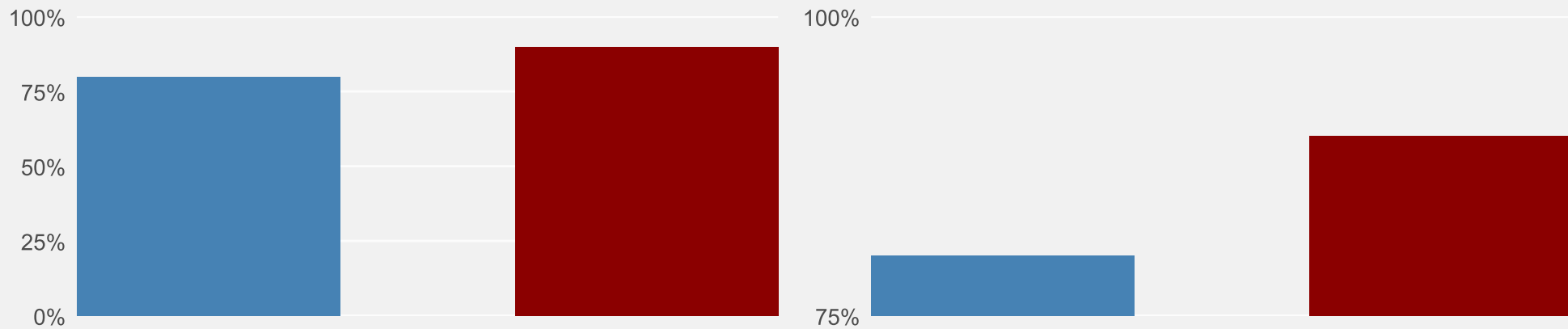
# Alternatives to bar plots

Raincloud plots show raw data, summary stats, and distribution



# Principle of proportional ink

Sizes of shaded areas should be **proportional** to the **data values** they represent

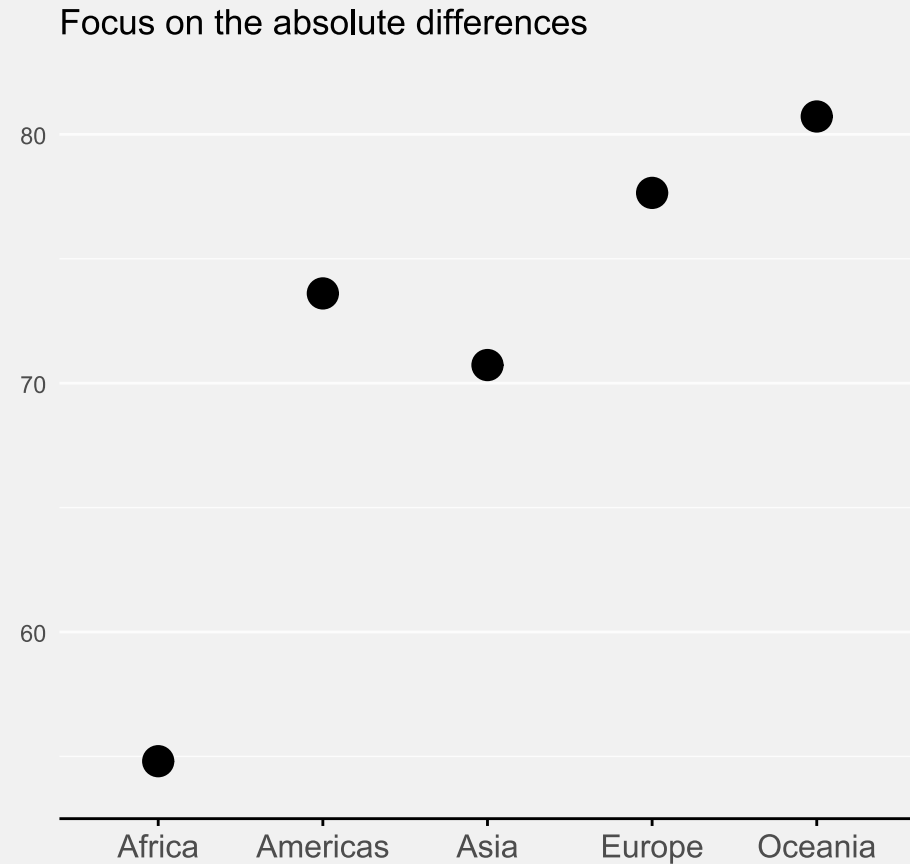
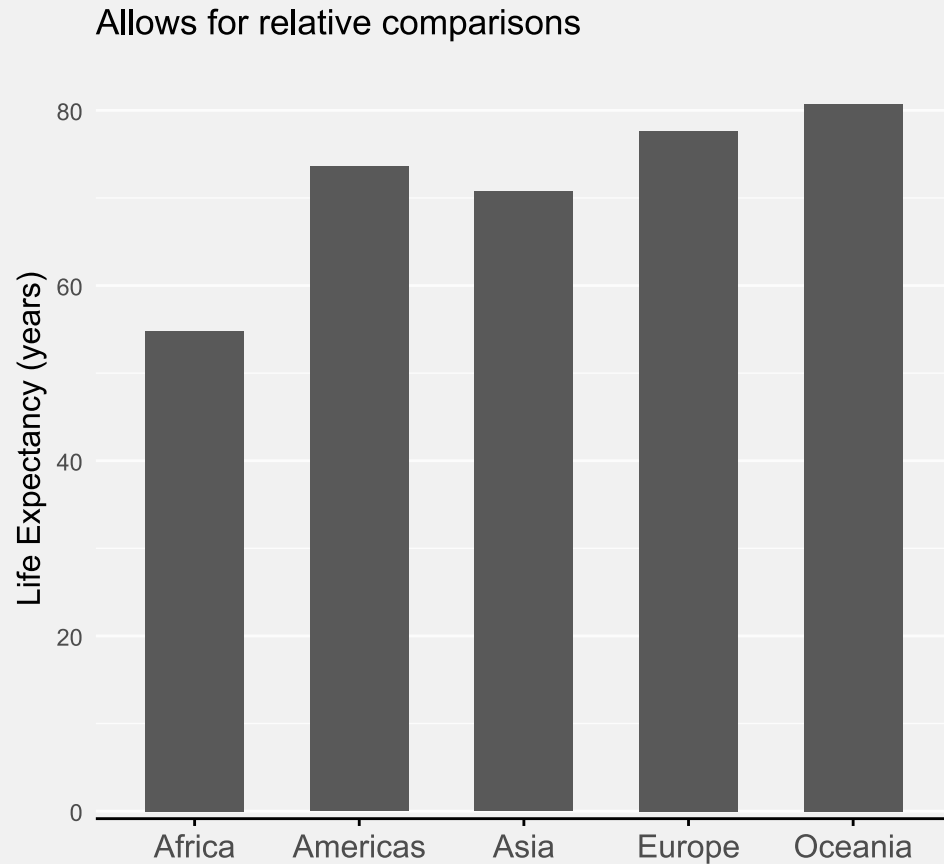


Here: bar length does not represent **relative data proportions** anymore

Always start bars at 0!

# Principle of proportional ink

Other plot types don't have to start at 0.

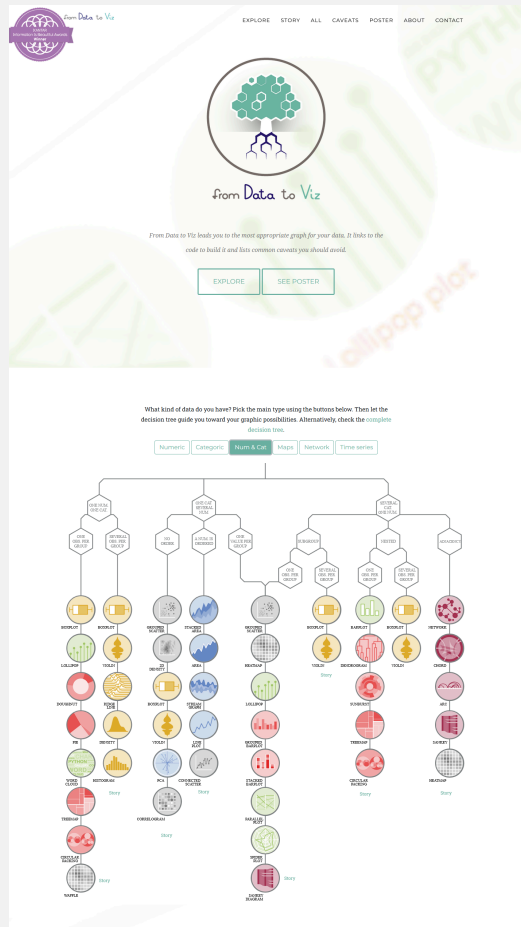


Data from the 'gapminder' R package

# 3: Choose the right chart type

# Choose the right chart type

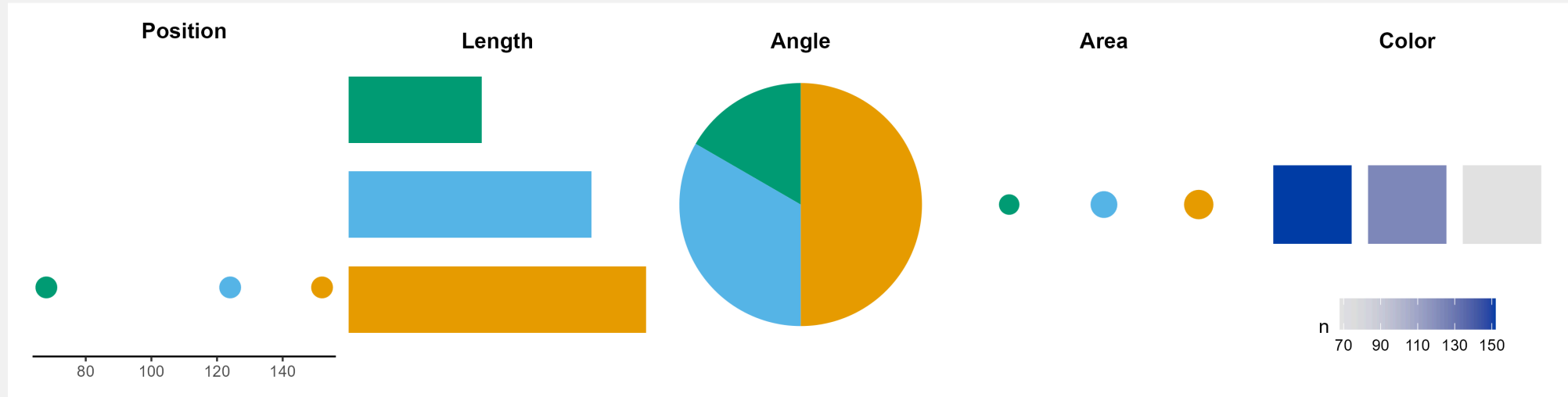
There are so many chart types - and cool tools to explore them



The dataviz project

From data vo viz

# Different channels - different accuracy



- Accuracy of judgement decreases from left to right
- More accurate judgements vs. more generic judgements

# Different channels - different accuracy

- Combine multiple channels for more accuracy
- Add numbers to increase accuracy of judgement



# 4. Focus on the core message

# Focus on the core message

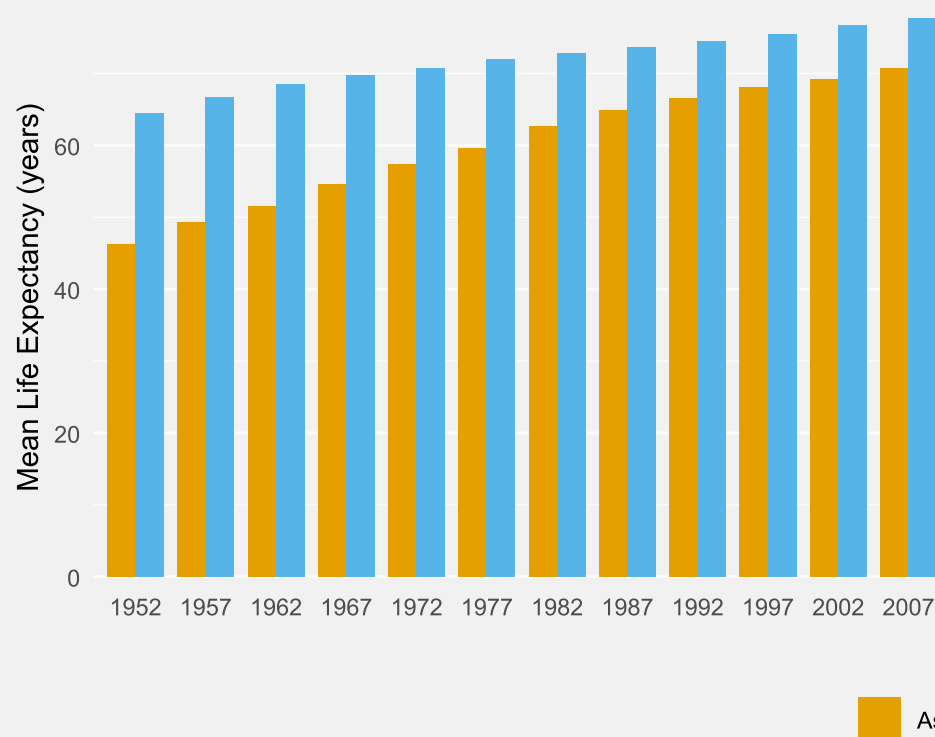
- The readers attention is limited: **be concise**
- Think about the **main message** you want to convey
  - Which variables do you need?
  - What can you omit?

# Arrange your plot

Arrange your plot so that it's **easy to extract** the main message

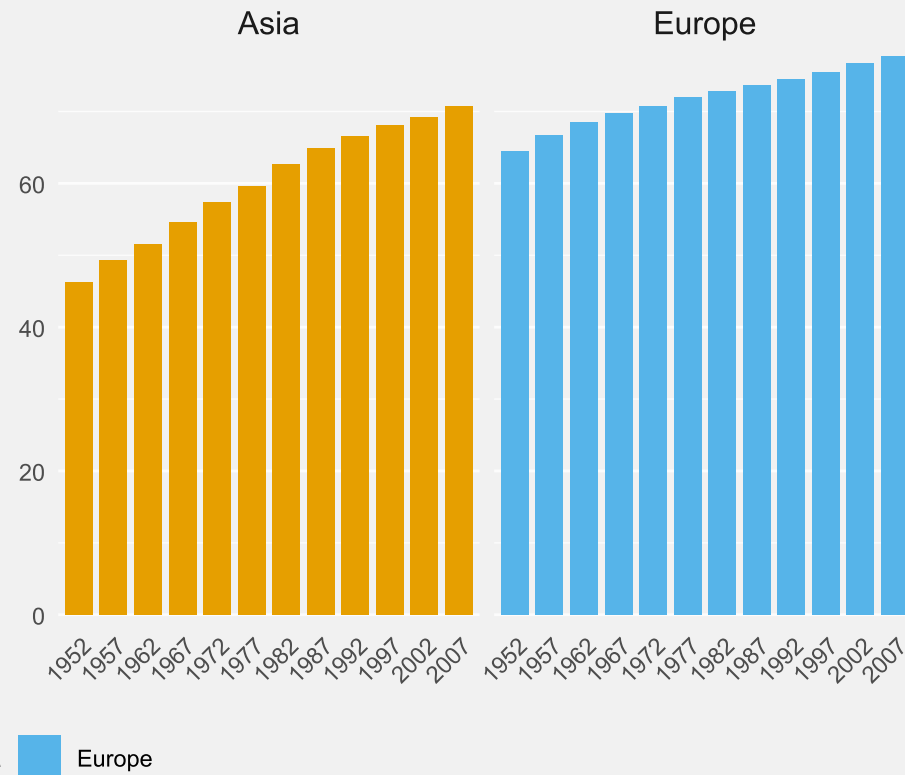
Life Expectancy group comparisons

Main message: Europe higher than Asia, Asia catching up



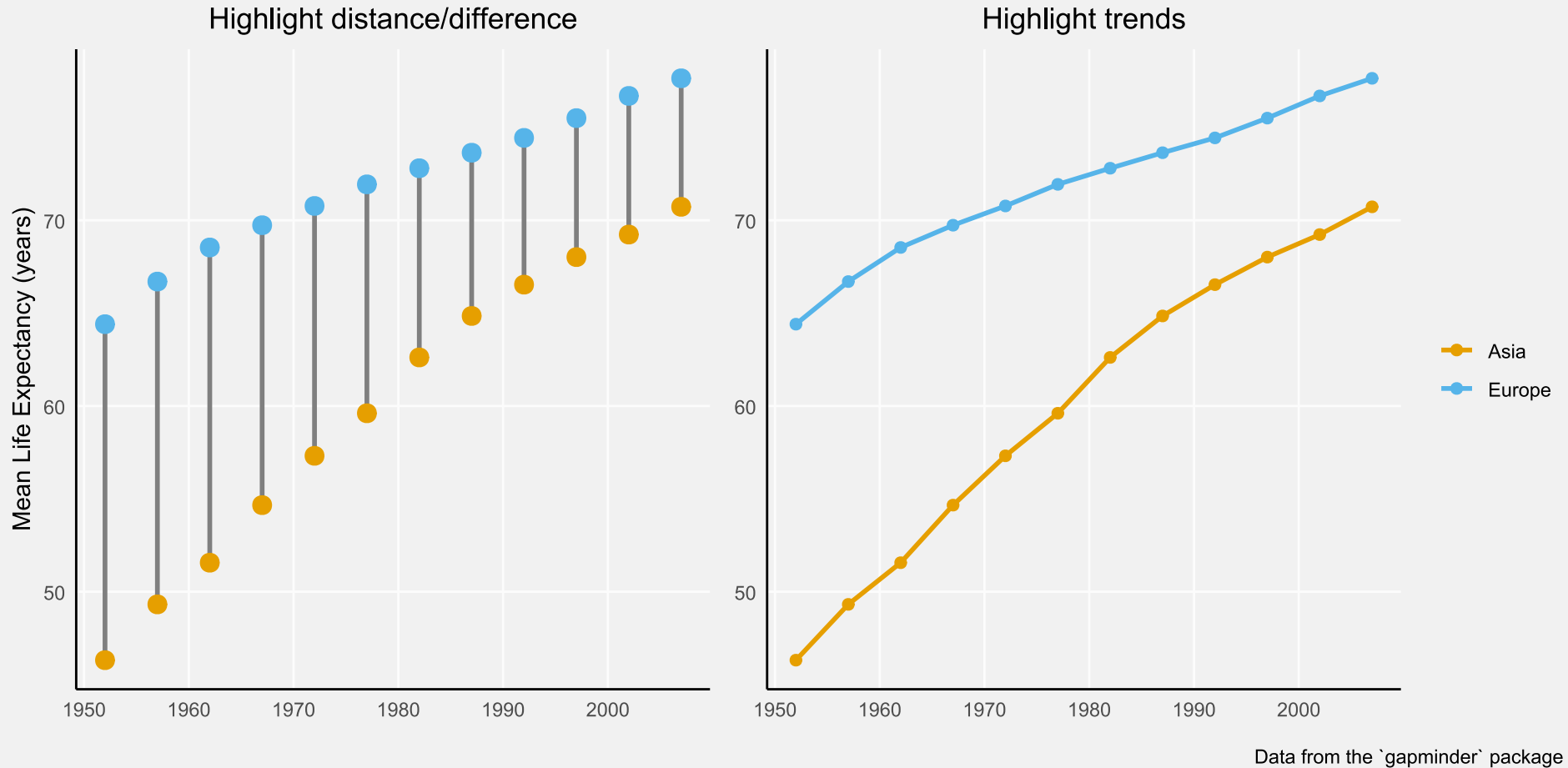
Life Expectancy Trends

Main message: Different rates of improvement over time



# Choose a good plot type

Different plot types tell different stories



# Keep it simple

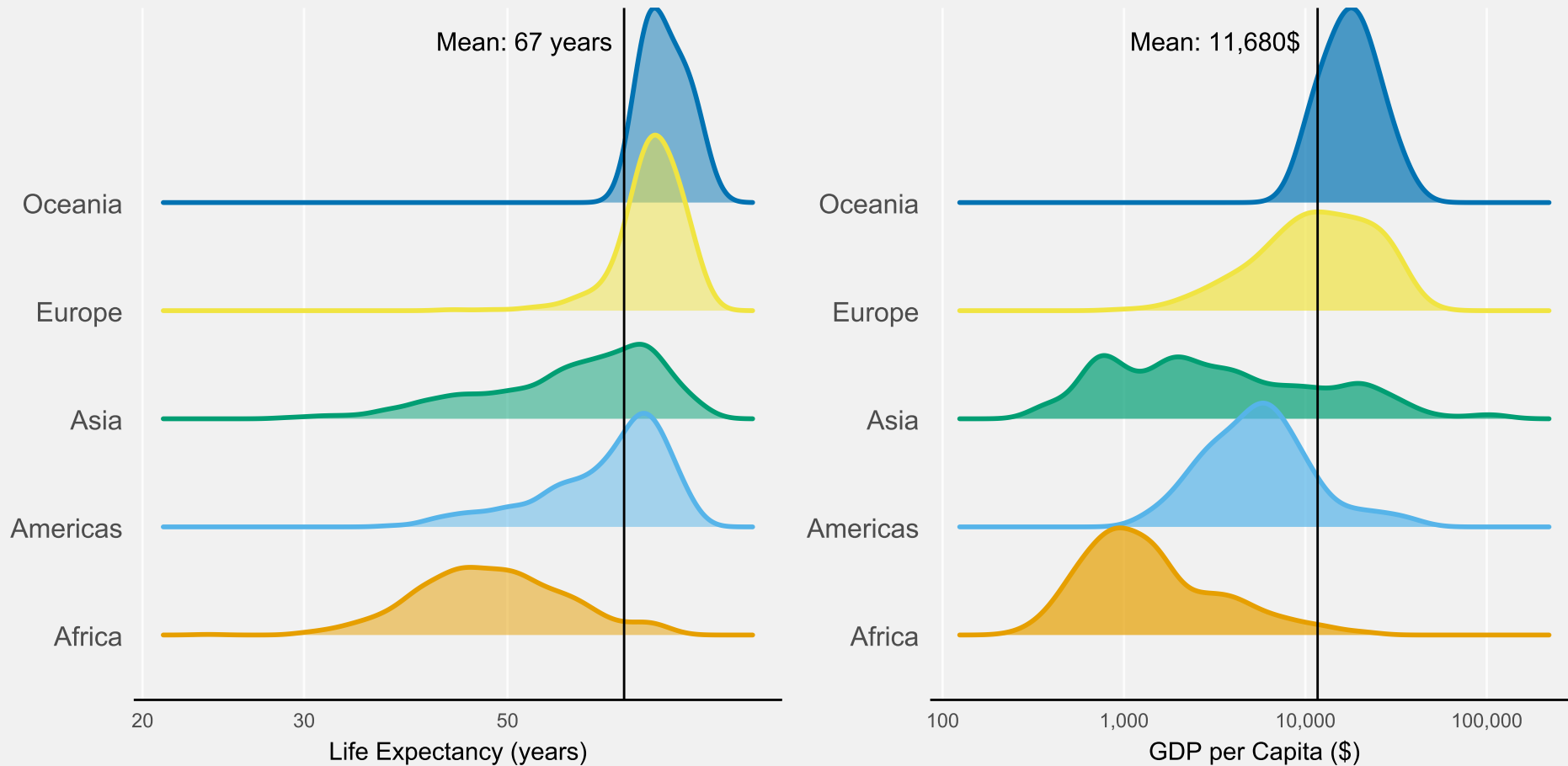
Don't overcomplicate your figures and bury your message

What is the main message here?



# Keep it simple

Message: Life expectancy and GDP differences in the world





## 5. Consider the trip

# Reading a figure is a timely experience

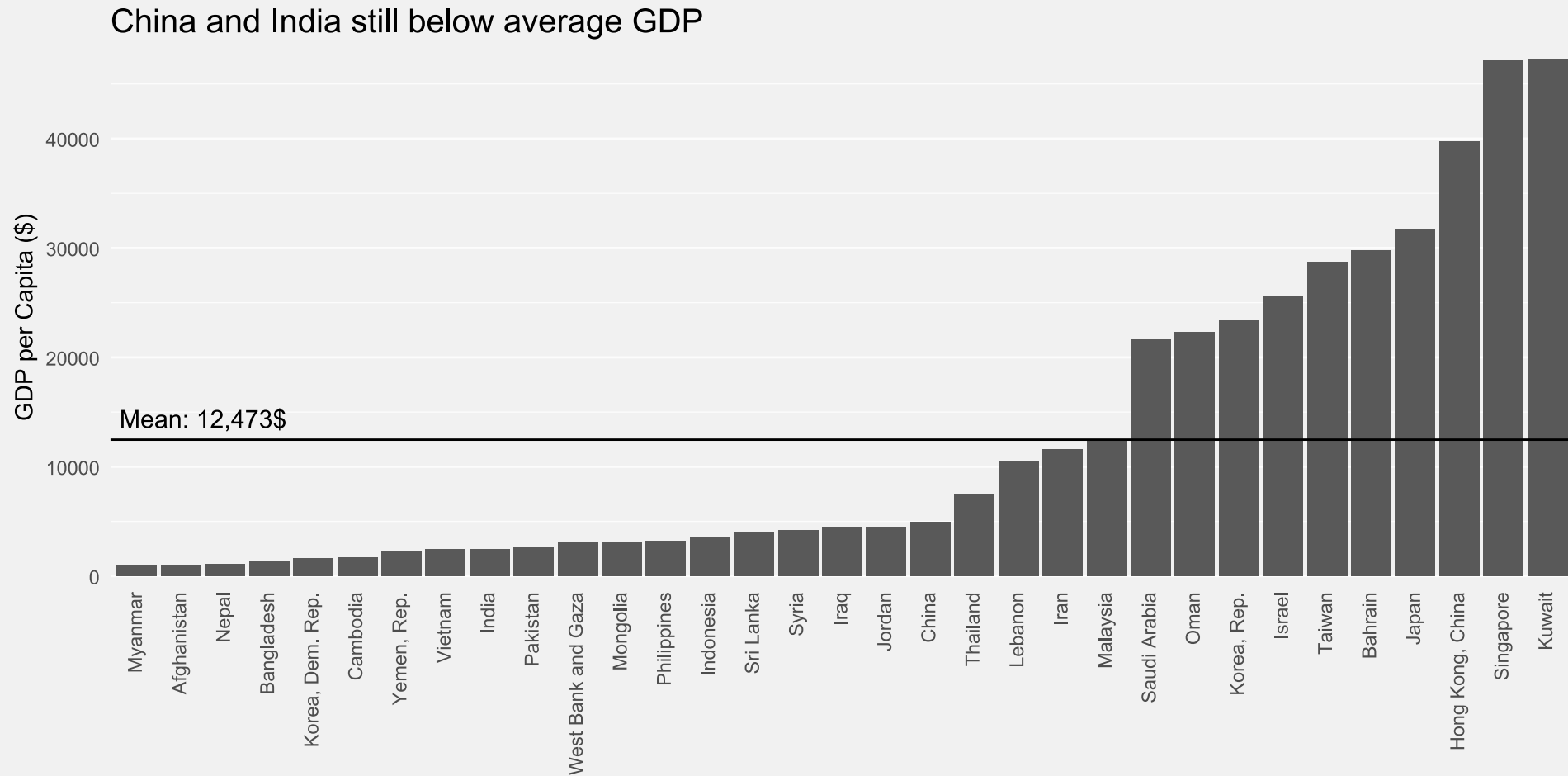
- We look at elements **step by step** before we come back to understand the **figure as a whole**
- Put yourself in the readers shoes
  - What will they look at first, second, etc.?
  - **How many steps** does it take to understand all the elements?

**Goal:** Make the trip (for the eyes and the brain) as short as possible



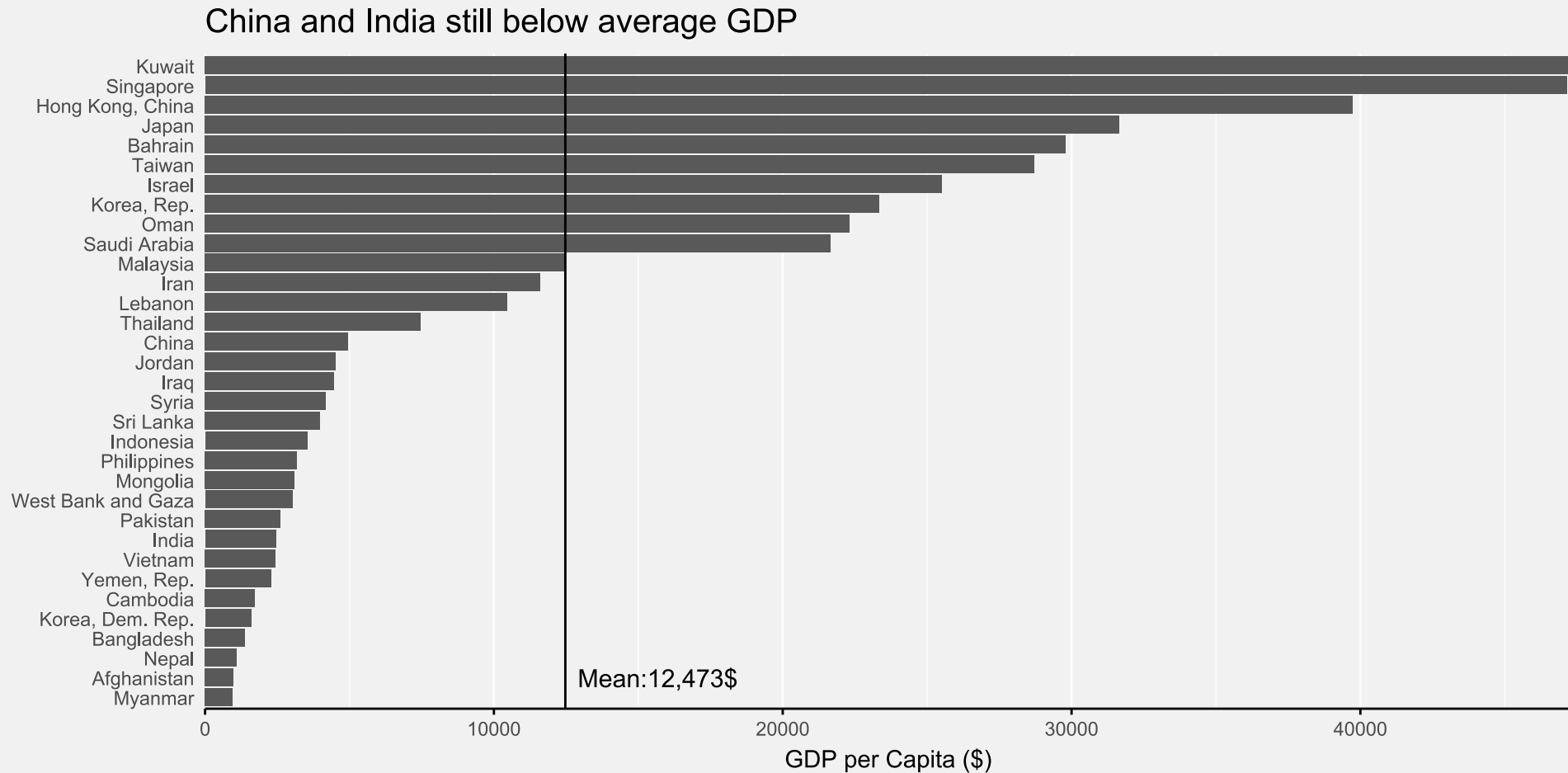
# Consider the trip

A story about the GDP China and India



# Rotate your plot

Reading labels upside down is a neck rotation - very annoying



# Highlight the main message

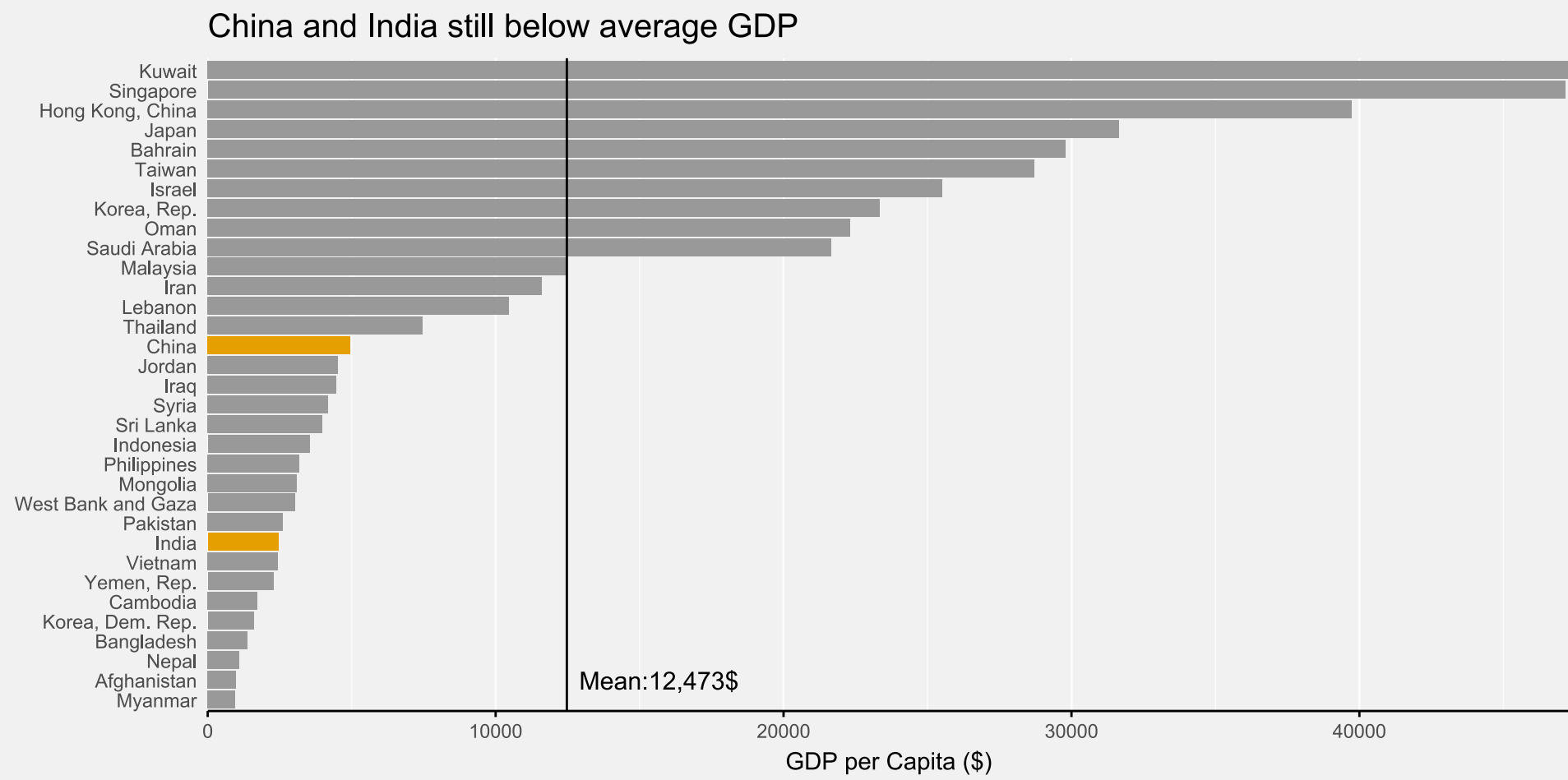
Use **highlighting** and de-emphasizing

Effective visualization helps us understand data quickly. **Patterns** emerge naturally, while **colors** enhance meaning. Good **design** choices and proper **emphasis** make insights accessible to everyone.

- Make use of **pre-attentive focus** (Things that *pop* out)
- Possible highlights: color, size, shape, arrows, ...

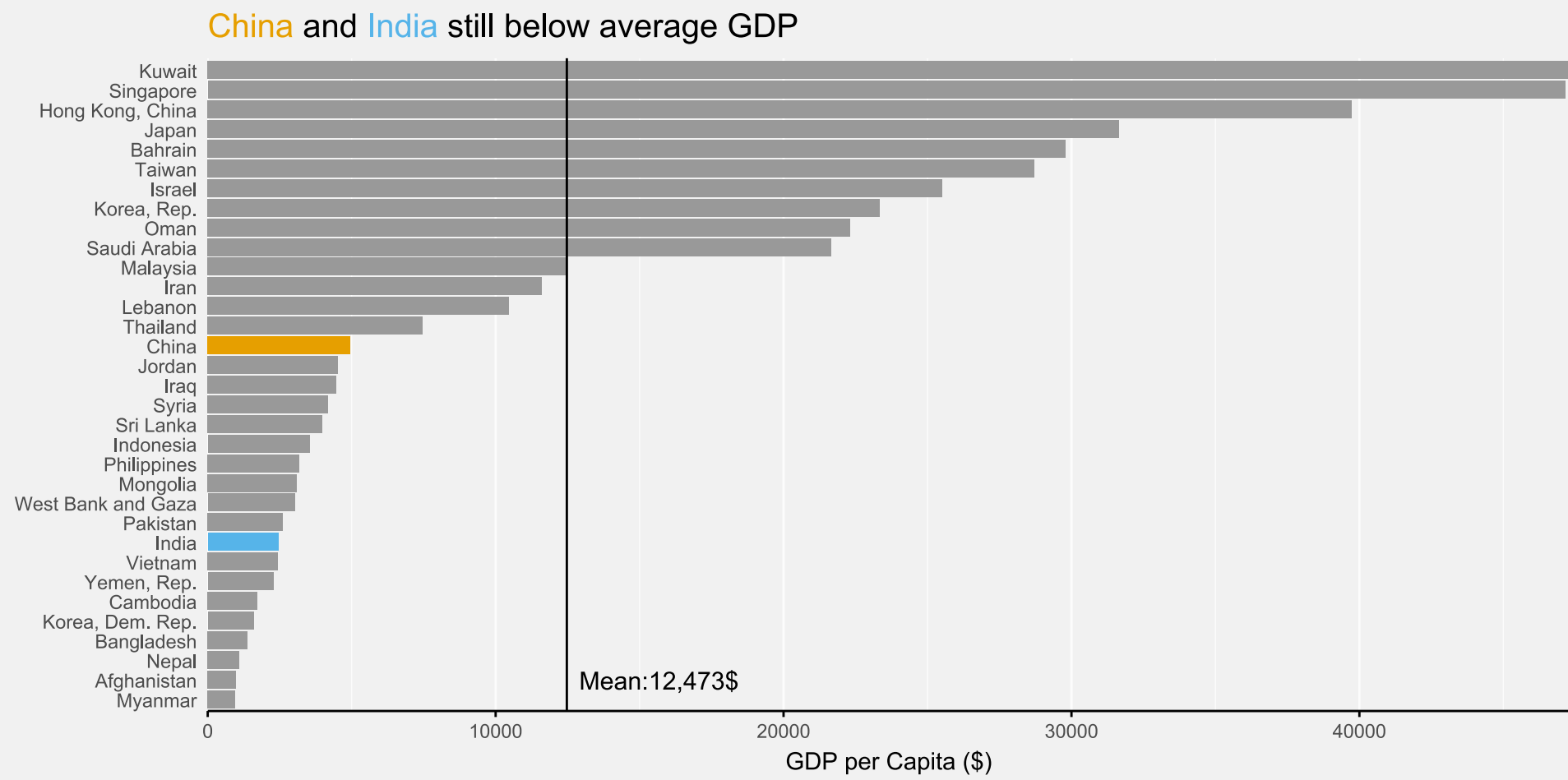
# Highlight the main message

Highlight focus contries, de-emphasize all others



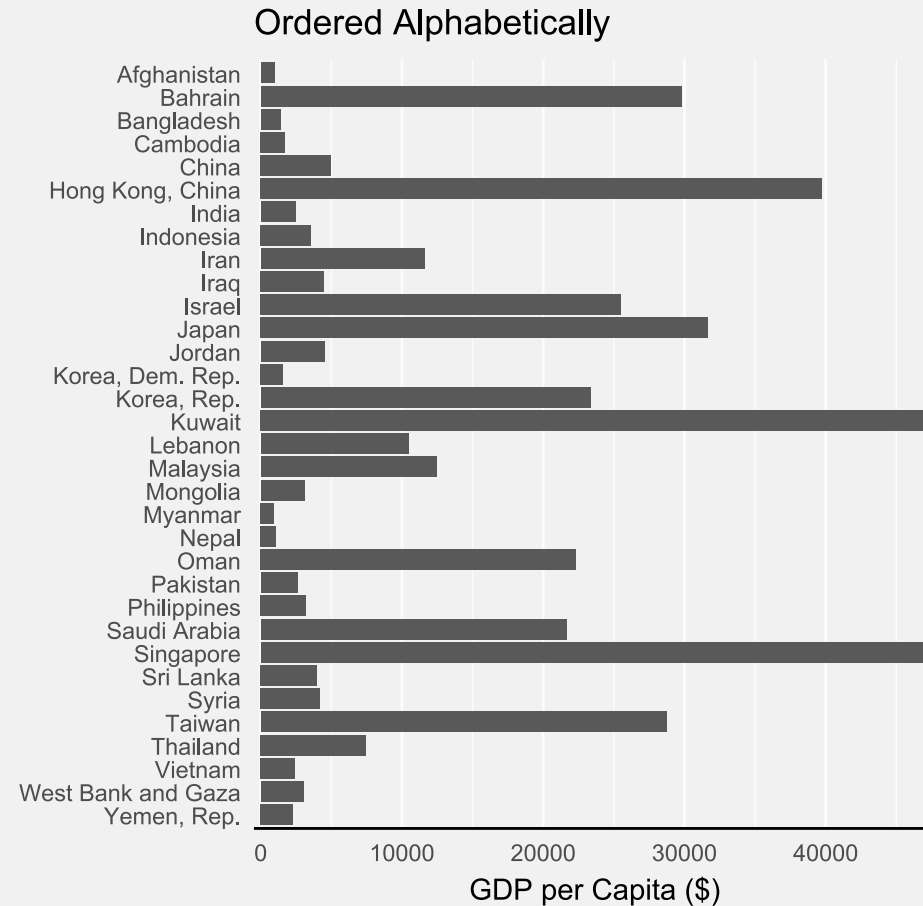
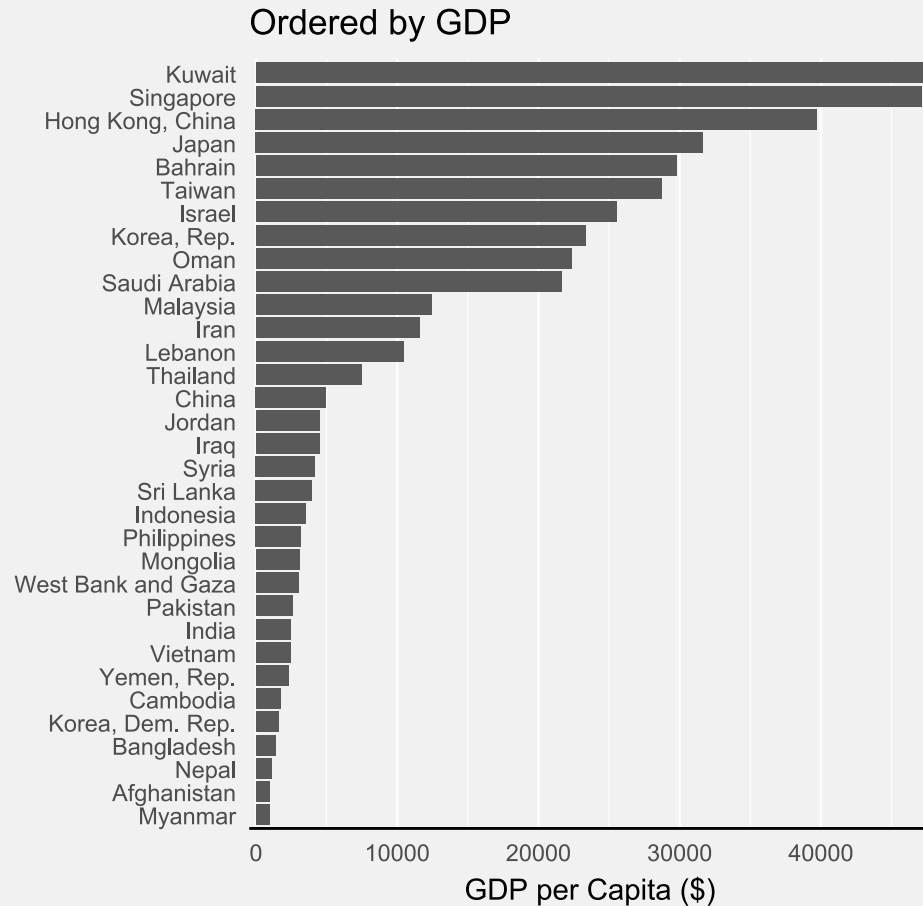
# Highlight the main message

Highlight focus contries, de-emphasize all others



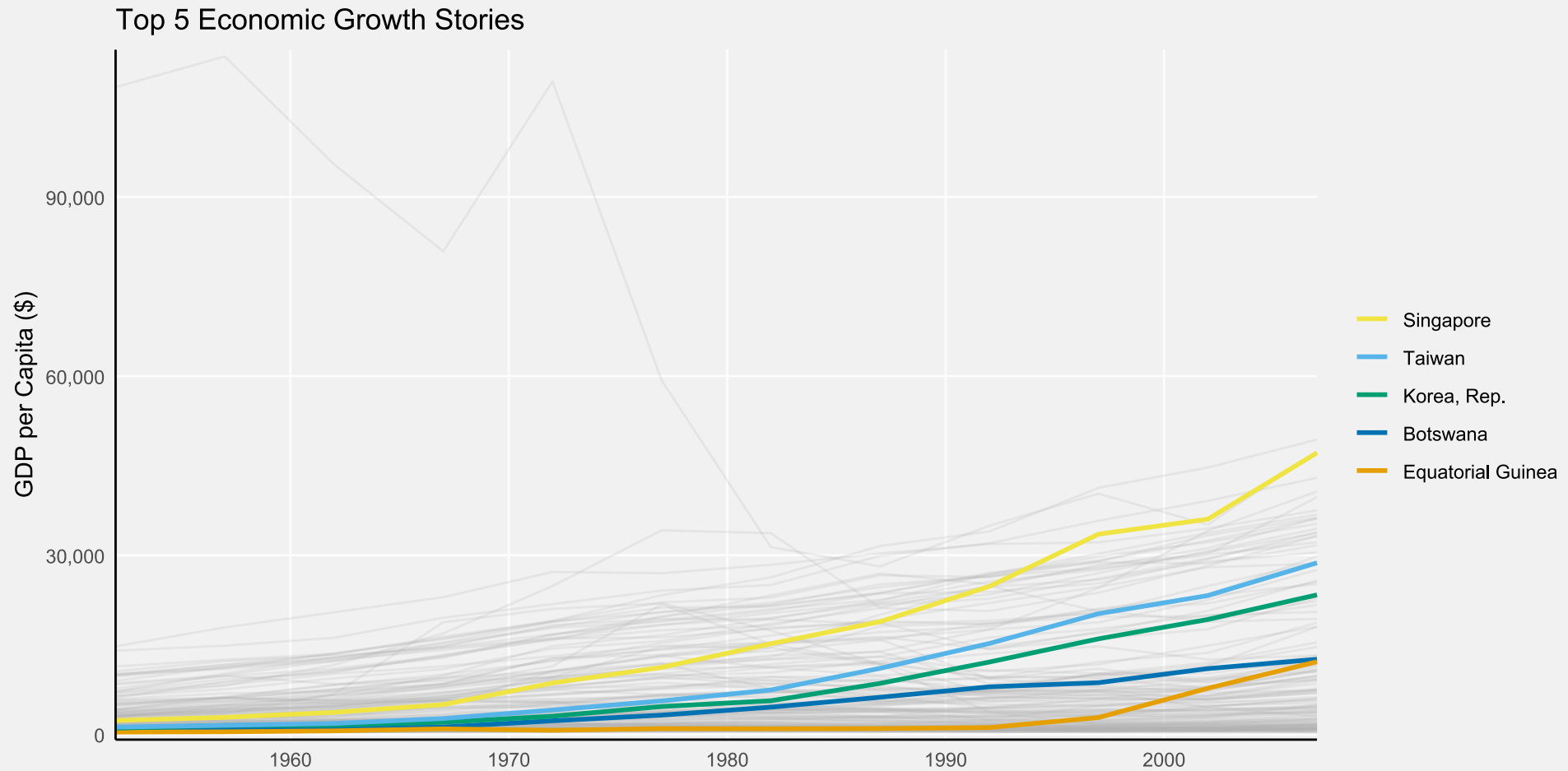
# Order your data

Order categories consciously not automatically.



# Order your data

Order categories consciously not automatically.





## 6. Less is more



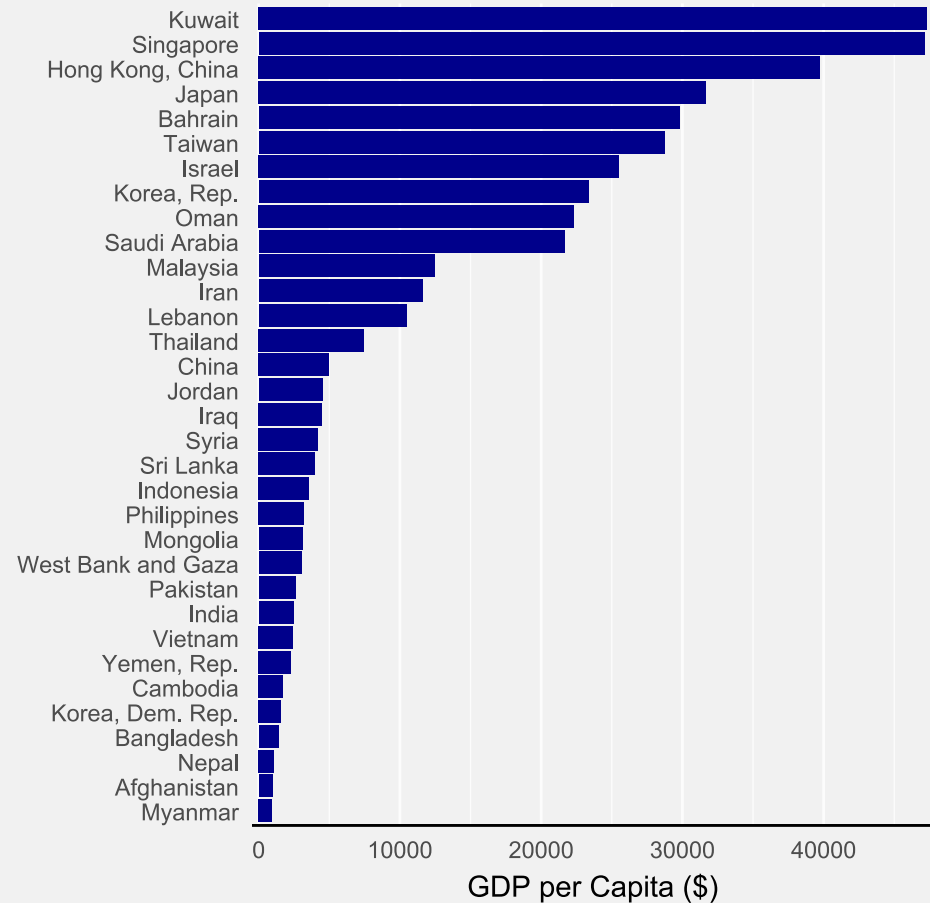
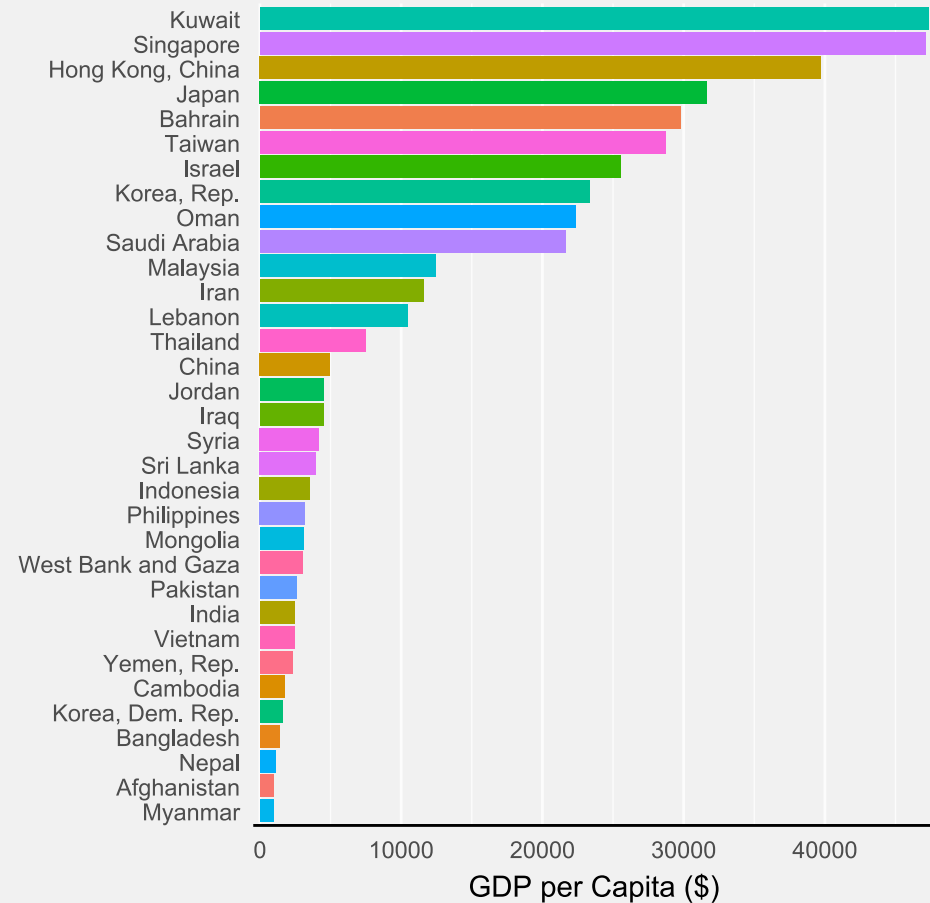
# The importance of differences

*Effective visualization helps us understand data quickly. Patterns emerge naturally, while colors enhance meaning. Good design choices and proper emphasis make insights accessible to everyone.*

Effective visualization helps us understand data quickly.  
**Patterns** emerge naturally, while **colors** enhance meaning.  
Good **design** choices and proper **emphasis** make insights accessible to everyone.

# The importance of differences

Use differences to communicate not to decorate

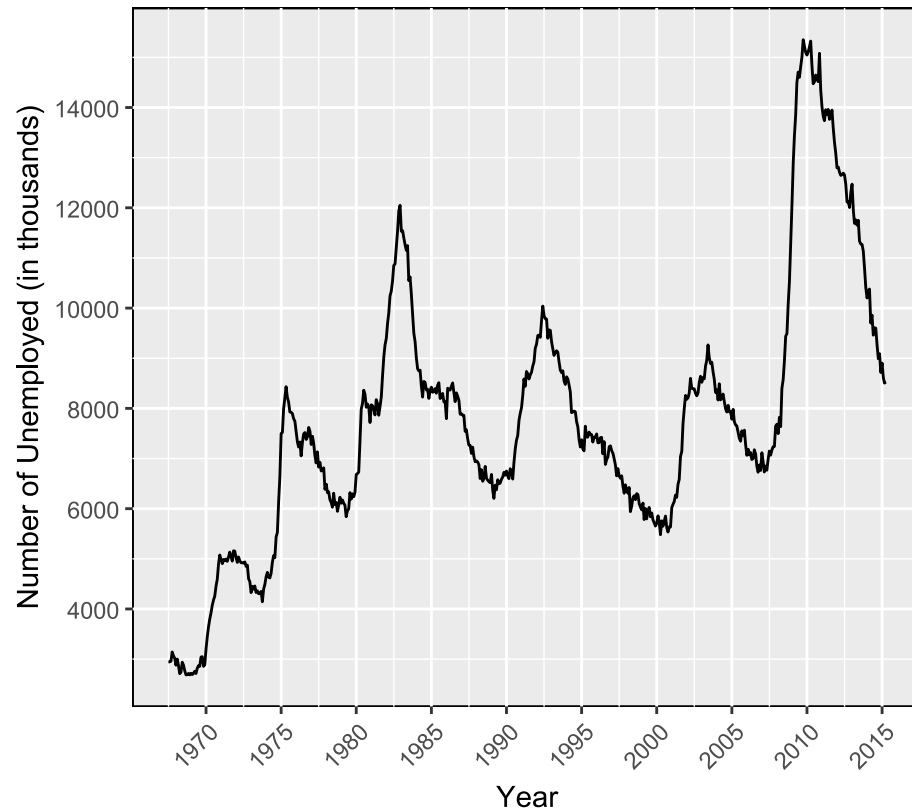


# Declutter your figure

- Try to maximize the **data/ink ratio**
- This is to an extent a matter of taste
- Remove redundant figure elements  
Excessive grid lines, boxes, duplicate text ...
- But keep elements important for reading

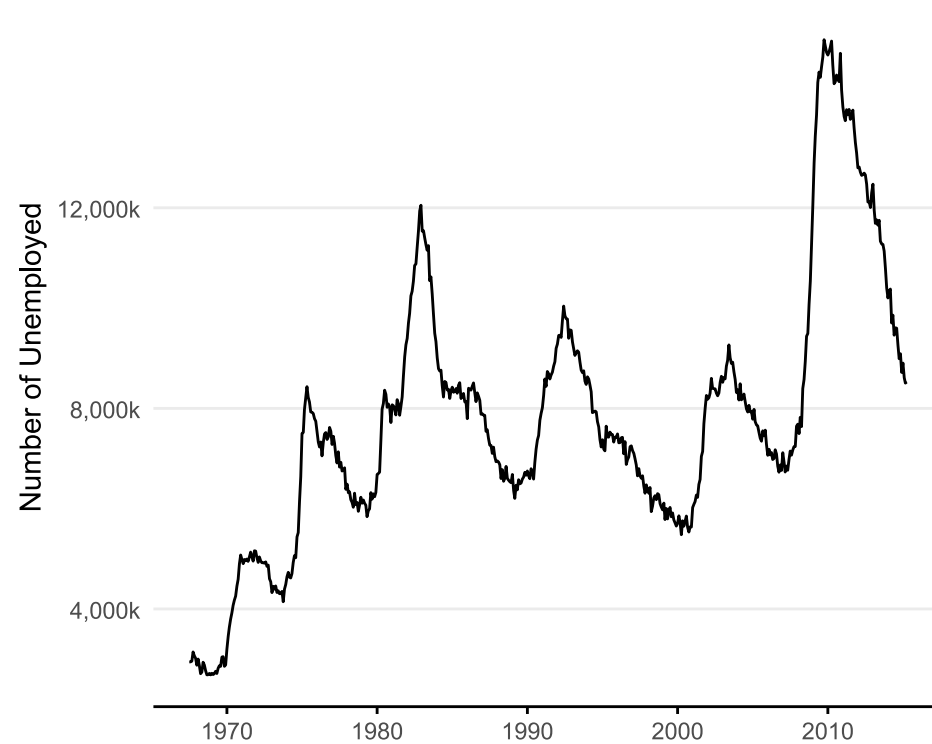
# Declutter your graphs

Time Series of Unemployment in the US



Data Source: Federal Reserve Economic Data (FRED)

Time Series of Unemployment in the US

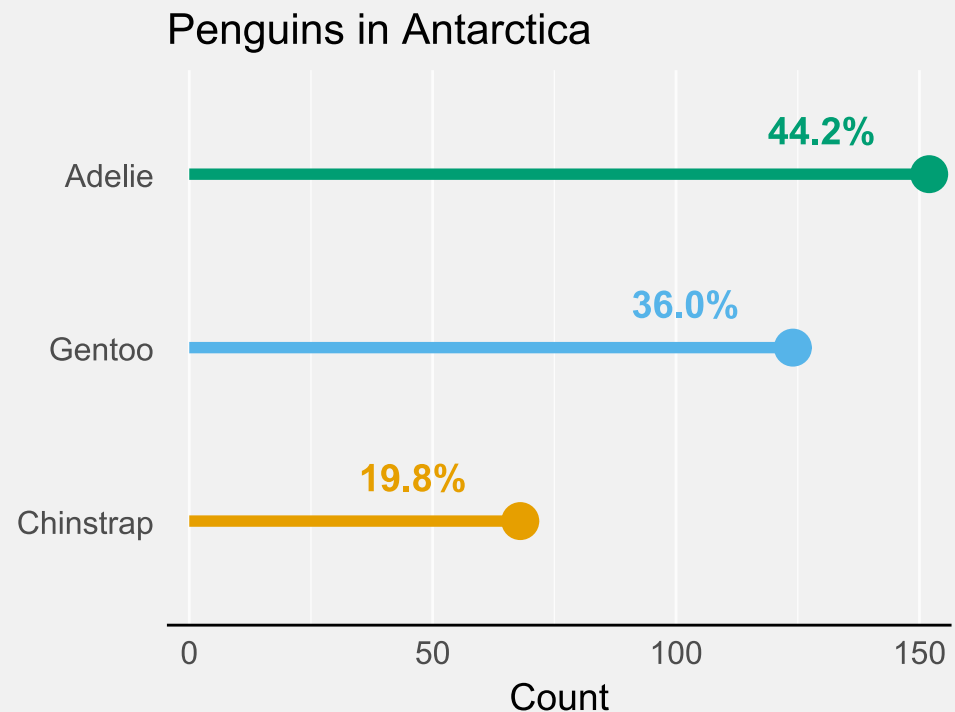
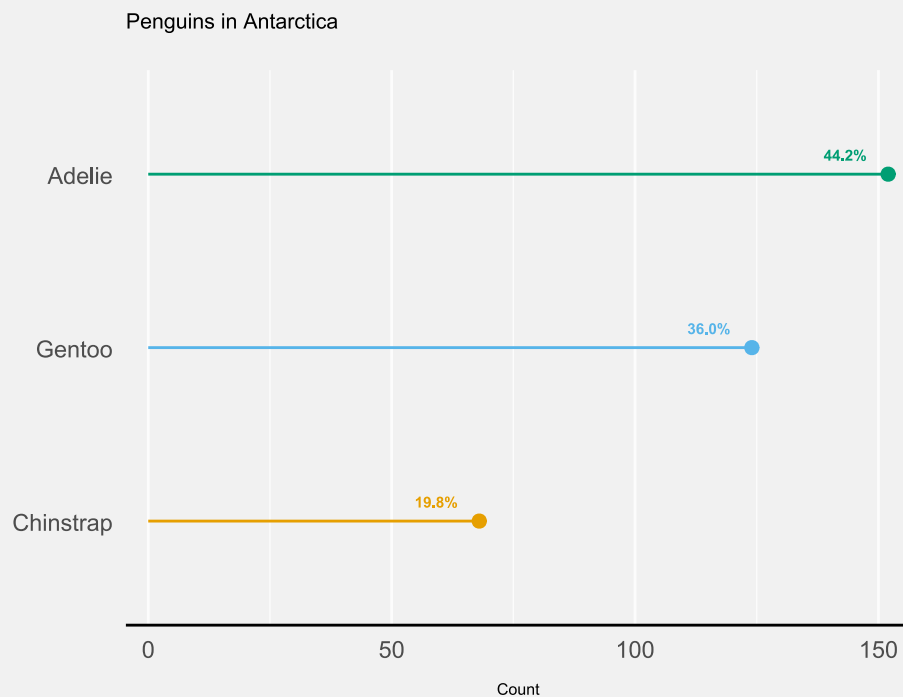


Data Source: Federal Reserve Economic Data (FRED)

# 7: Make it accessible

# Element size

- Make sure your elements are big enough  
Text size, Linewidth, Point size
- Depends on the context



# Contrast

Make sure that the contrast is high enough

|       |                      |                      |             |             |
|-------|----------------------|----------------------|-------------|-------------|
| 1:1   |                      |                      |             |             |
| 1.1:1 | You dislike readers. | That's bad.          | Nope        | Nope        |
| 1.5:1 | Not ideal.           | Not ideal.           | That's bad. | That's bad. |
| 3:1   | Can be ok.           | Can be ok.           | Not ideal.  | Not ideal.  |
| 4.5:1 | Safe for large text. | Safe for large text. | Ok.         | Ok.         |
| 7:1   | Safe.                | Safe.                |             |             |

Blogpost on colors by Lisa Charlotte Muth (Datawrapper)

- Use tools to check contrast, e.g.  
[https://snook.ca/technical/colour\\_contrast](https://snook.ca/technical/colour_contrast)

# Color

Use logical/intuitive colors



Blogpost on colors by Lisa Charlotte Muth (Datawrapper)



# Color

Choose colorblind friendly palettes (if in doubt: test!).

## VIZ PALETTE

By: [Elijah Meeks](#) & [Susie Lu](#)

### PICK

Use Chroma.js

Use ColorGorical

Use ColorBrewer

Add Replace

### EDIT

7 Colors

Add

☒ #hex ☐ rgb ☐ hsl

|   |   |         |   |
|---|---|---------|---|
| 1 | ● | #ffd700 | ✕ |
| 2 | ● | #ffb14e | ✕ |
| 3 | ● | #fa8775 | ✕ |
| 4 | ● | #ea5f94 | ✕ |
| 5 | ● | #cd34b5 | ✕ |
| 6 | ● | #9d02d7 | ✕ |
| 7 | ● | #0000ff | ✕ |

### GET

☒ String quotes  
☐ Object with metadata

```
[{"#ffd700",  
"#ffb14e",  
"#fa8775",  
"#ea5f94",  
"#cd34b5",  
"#9d02d7",  
"#0000ff"}]
```

## COLORS IN ACTION

Background color: #ffffff  
Font color: ● #000000  
Charts made with [Semiotic](#)

**Color Population:**

|                           |                      |                     |                    |                      |
|---------------------------|----------------------|---------------------|--------------------|----------------------|
| No Color Deficiency - 96% | Deuteranomaly - 2.7% | Protanomaly - 0.66% | Protanopia - 0.59% | Deuteranopia - 0.56% |
|---------------------------|----------------------|---------------------|--------------------|----------------------|

Greyscale

Sample font Randomize Data Stroke: Dark None

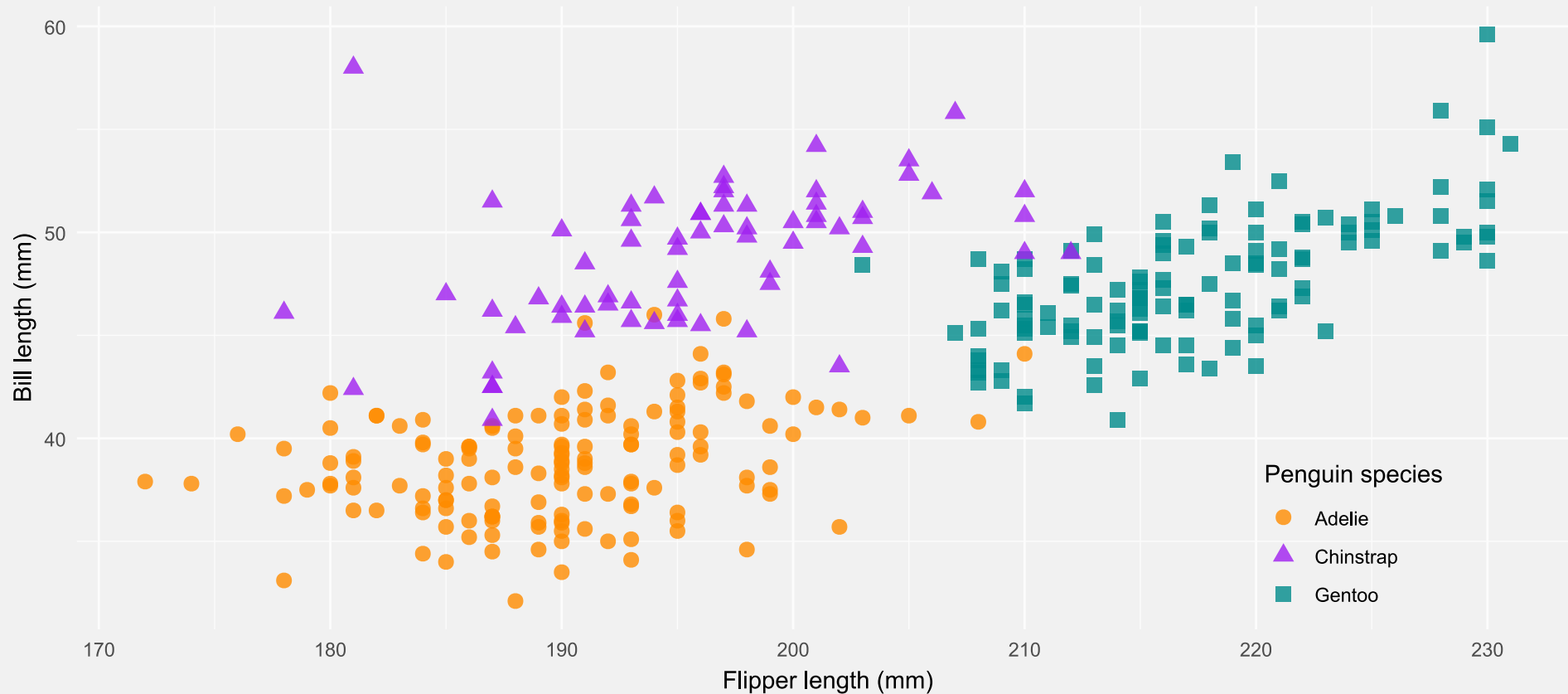
Use the [Viz Palette tool](#) to test for color blindness

# Add redundancy

Redundancy increases the chance that everyone can see the difference!

## Flipper and bill length

Dimensions for Adelie, Chinstrap and Gentoo Penguins at Palmer Station LTER



# Summary

1. Consider the **context**
2. Make your data **transparent**
3. Choose the **right chart type**
4. Focus on the **core message**
5. Consider the **trip**
6. **Less is more**
7. Make it **accessible**

Start analyzing these points in yours and other people's plots.

# Next lecture

Topic t.b.a.

 15th May  3 - 4 p.m.  Webex

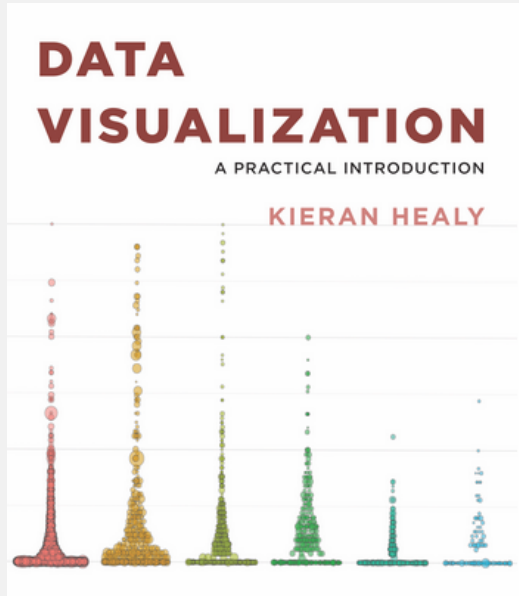
 Subscribe to the mailing list

 For topic suggestions and/or feedback [send me an email](#)

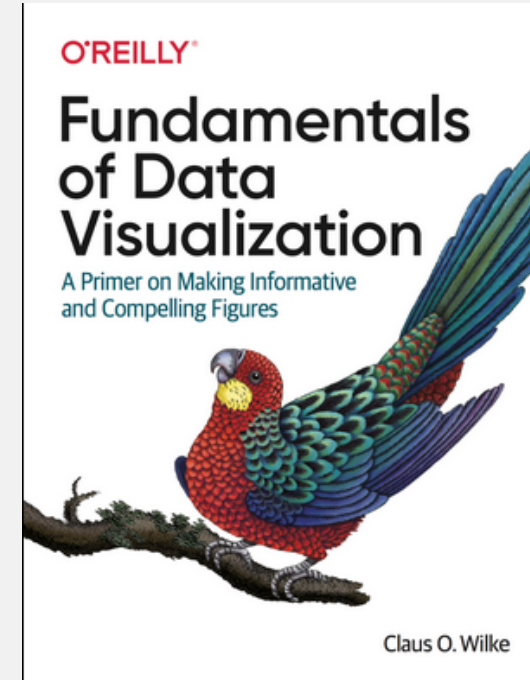
# Thank you for your attention :)

Questions?

# References



Healy, K. (2018). Data Visualization: A Practical Introduction. Princeton University Press.



Wilke, C. O. (2019). Fundamentals of Data Visualization. O'Reilly Media.